SUMITOMO ELECTRIC LIGHTWAVE

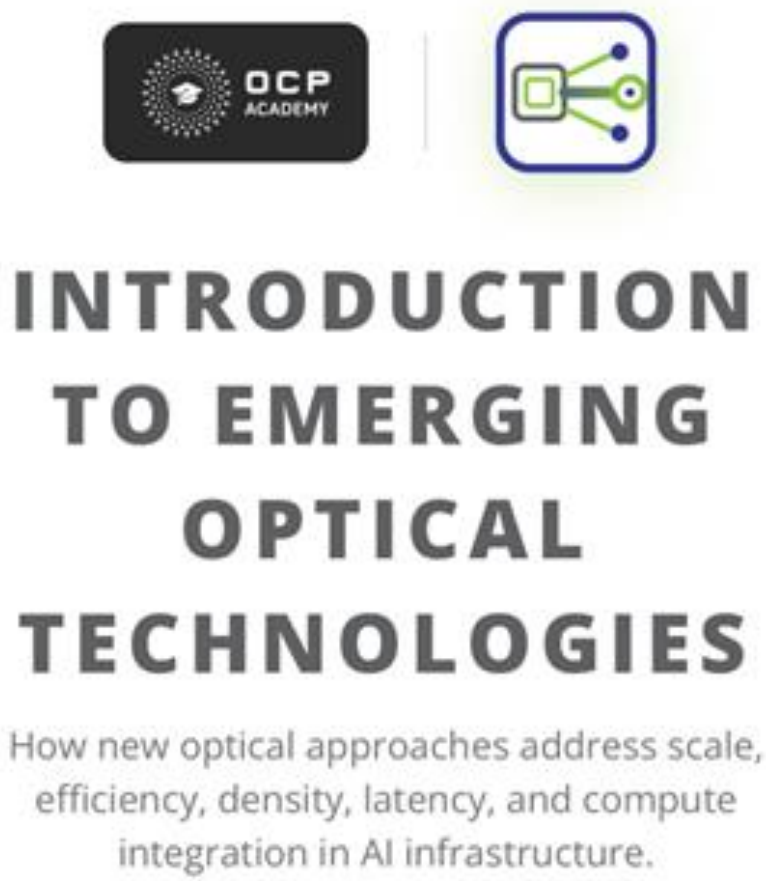
MULTI-CORE FIBER

A Solution for Scale Up, Out, and Across

Dwayne Holst, Optical Communications Engineer

Thank you to OCP Academy

- Big Thanks to James Kelly
- Senko and Viavi



PROBLEM STATEMENT

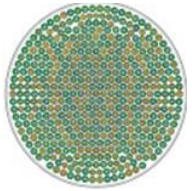
GPU Generation Progression Across Exponential Fiber Density Demands

Hopper –
H200 GPUs (Hopper)

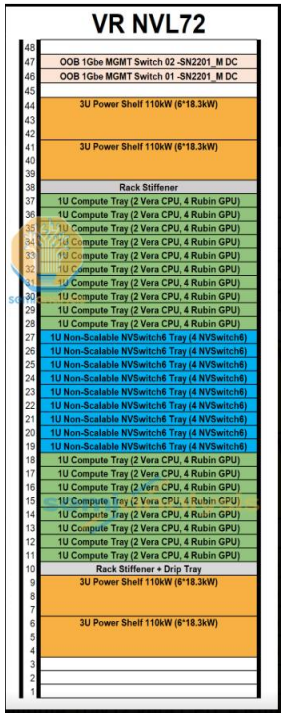


Specifications:
4 sleds x 8 GPUs
per node =
32 GPU dies per rack

Total BE Scale Out Fibers
Per GPU Rack: **~576f BE +**
64f FE



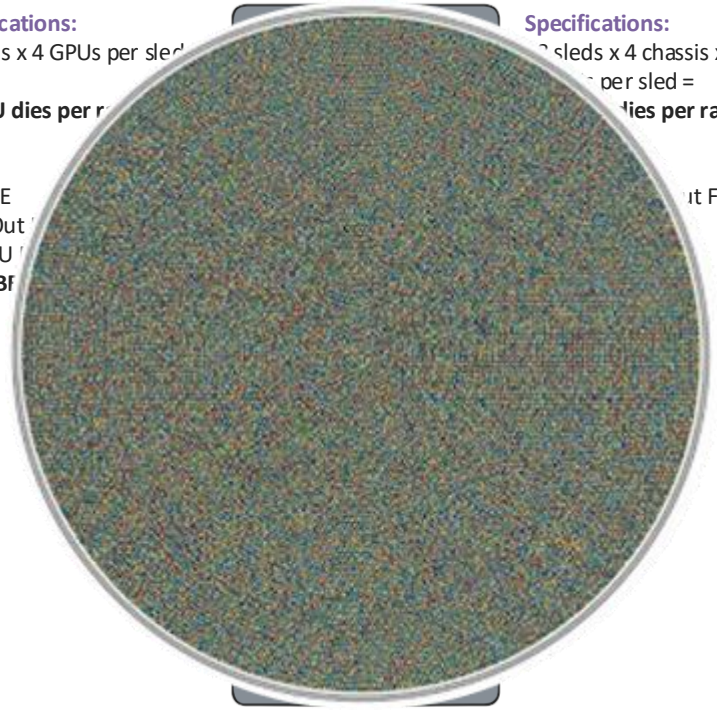
Blackwell –
GB200 GPUs (Oberon)



Specifications:
18 sleds x 4 GPUs per sled
=
72 GPU dies per r

Total BE
Scale Out
Per GPU
2308f BE

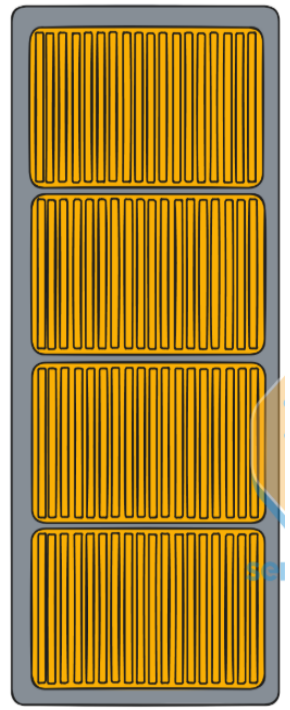
Rubin –
VR300 GPUs (Kyber)



Specifications:
sleds x 4 chassis x
per sled =
dies per rack

ut Fibers Per

Feynman –
F300 GPUs (Kyber+)



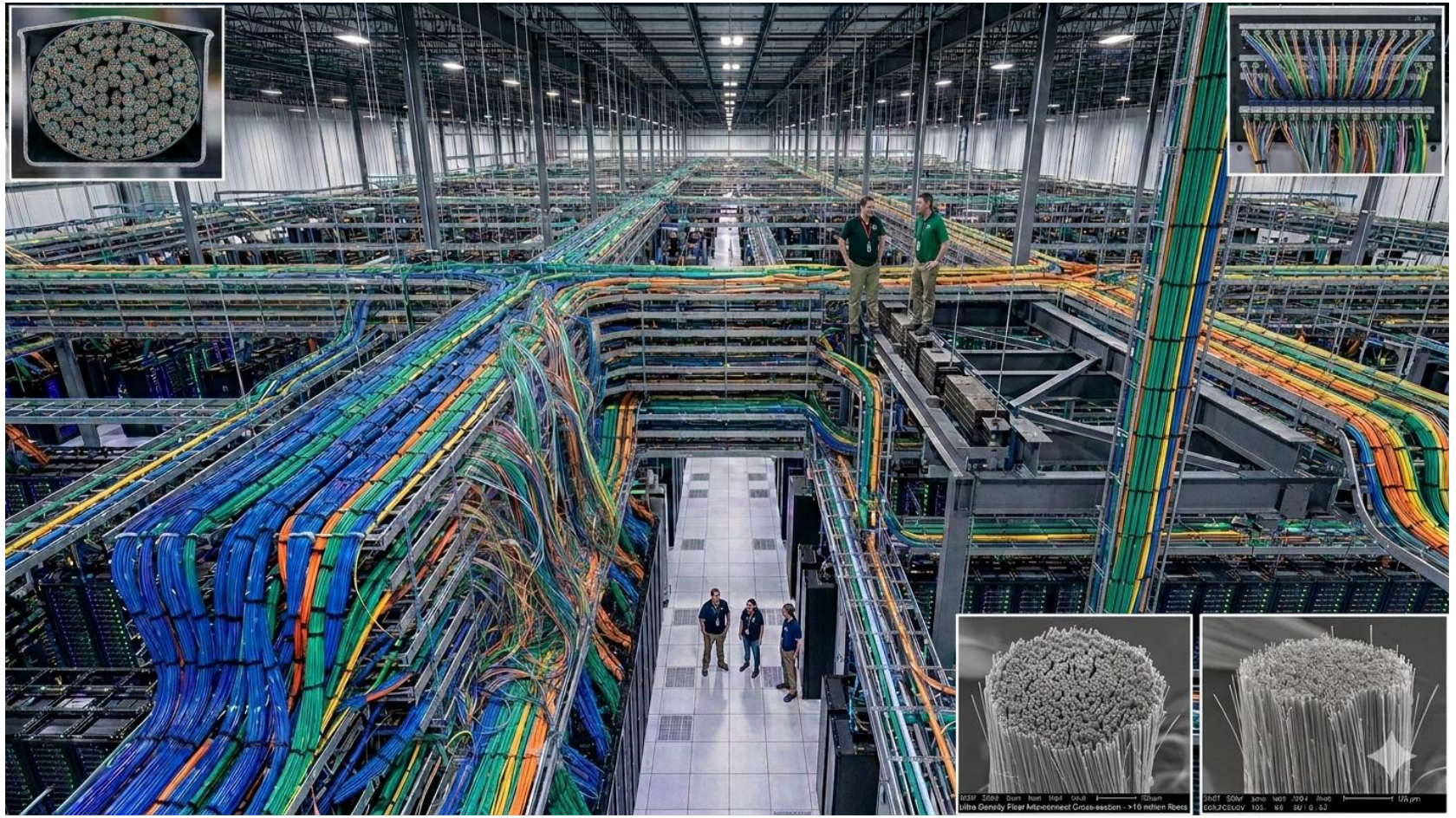
Specifications:
18 sleds x 4 chassis x
4 GPUs per sled =
576 GPU dies per rack

Total BE Scale Out Fibers Per
GPU Rack:
18,464 Fibers!!!

2021-2023 | 2024-2026 | 2027-2029 | 2030+

Where Today's Infrastructure Breaks

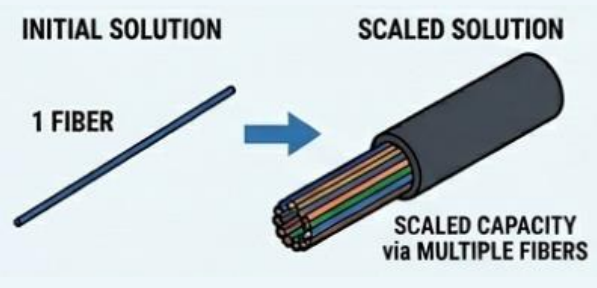
Micro-bending losses
Load-bearing limitations
Thermal aggregation
Routing complexity
Maintenance and scalability



SOLUTION SET OVERVIEW

Parallel Optics vs. WDM vs. SDM

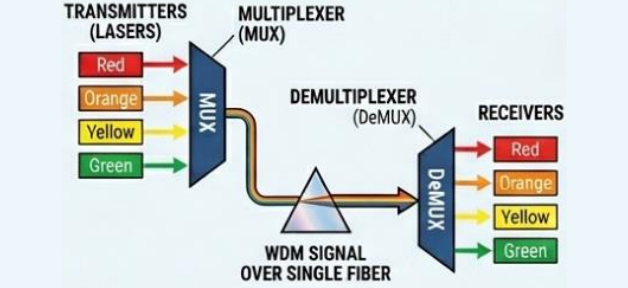
SCALING VIA ADDING FIBERS



Parallel Optics Scaling

Today's most common solution (MPO/MMC), however it is approaching practical limit

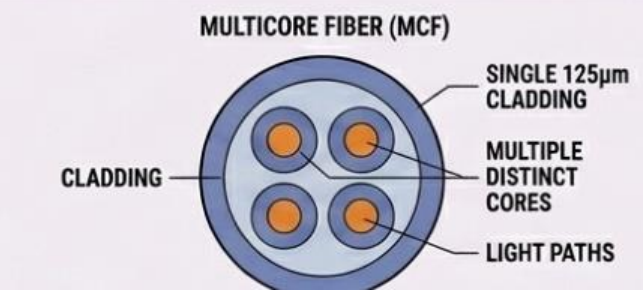
SCALING VIA SPECTRUM



Wave Division Multiplexing (WDM)

Not ideal for sub 500m links (added power, cost, and latency)

SCALING VIA CORES



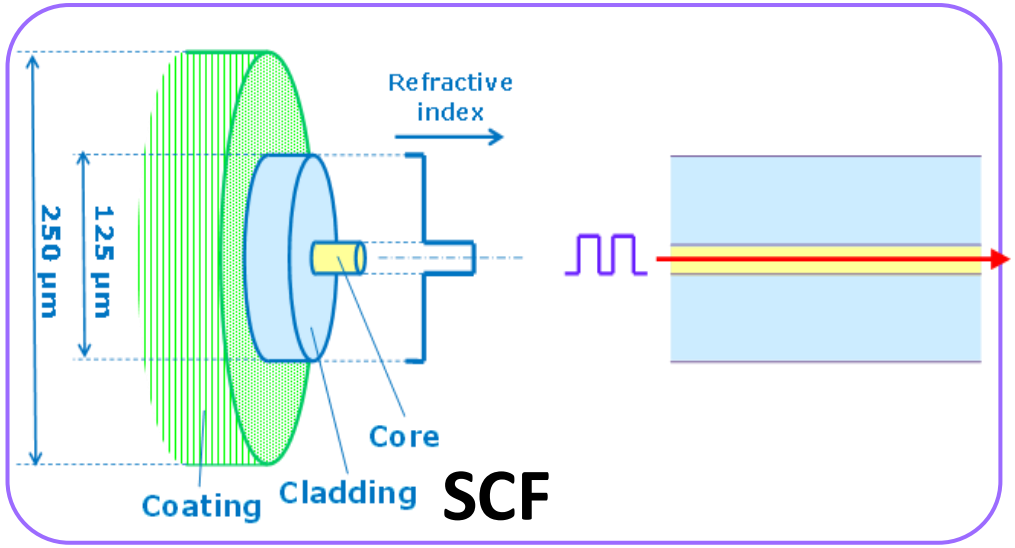
Spatial Division Multiplexing (SDM)

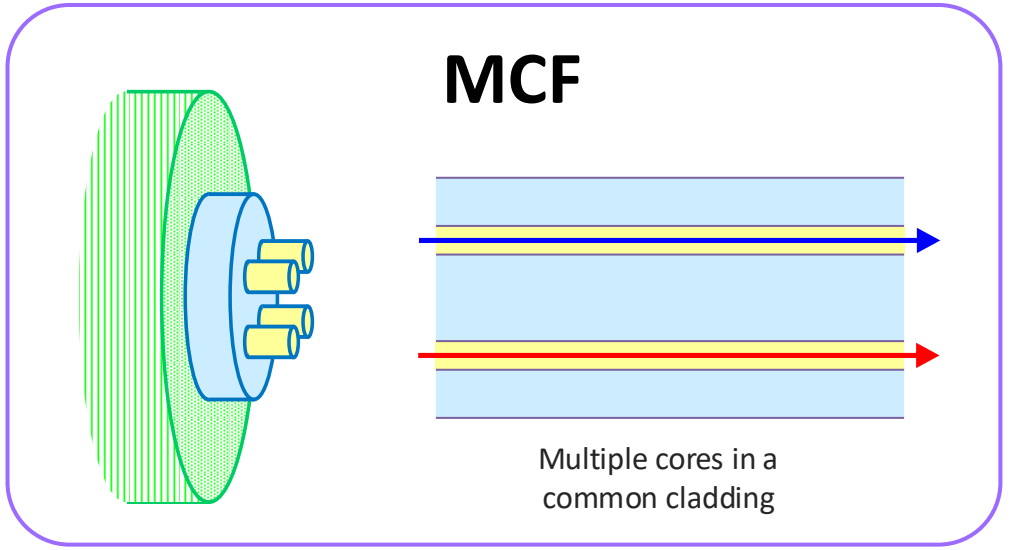
Leading SDM solution Multi-core fiber – ideal for saving space, weight, installation time

Multi-Core Fiber Solutions

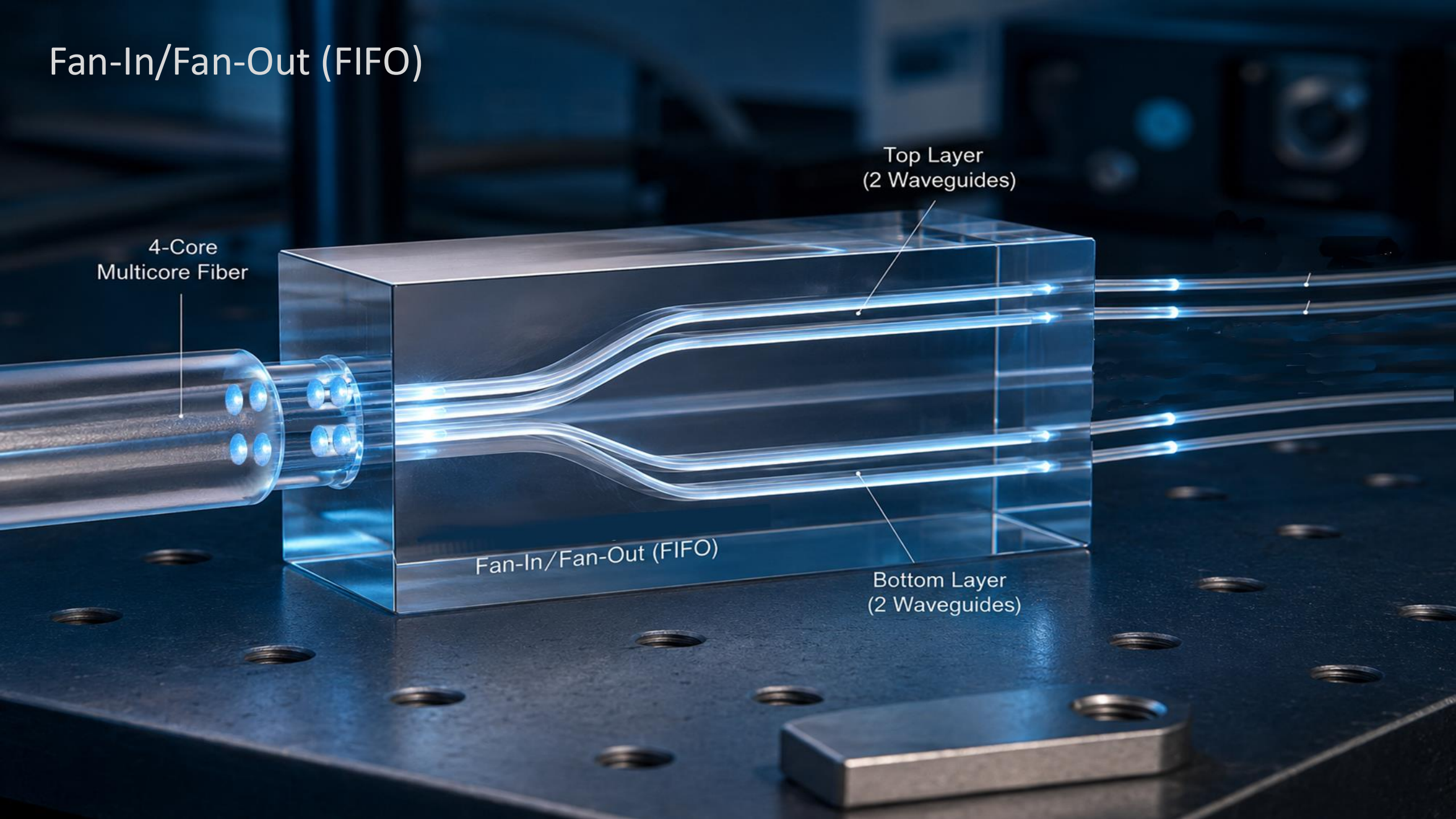
In response to demands for scalable and cost-effective fiber cable infrastructure for data center networks, using space-division multiplexing (SDM) to achieve cost-efficient network capacity scaling, Sumitomo Electric Industries (SEI) has developed a line of multi-core fibers (MCFs) and connectivity products offering a viable approach to increase the relative **core density** in a given system, easing common installation problems of very large fiber counts in data center links.

- Increase fiber density
- Increase network bandwidth
- Maximize existing conduit/cable tray/rack space
- Preserve insertion loss levels
- Maintain small form factors
- Limit channel crosstalk
- **MSA Agreement:** Agreement reached between major producers of MCF, standardizing on 4-core MCF design requirements





Fan-In/Fan-Out (FIFO)



Multi-Core Fiber – Space & Weight Savings

Higher Transmission Capacity in Limited Space

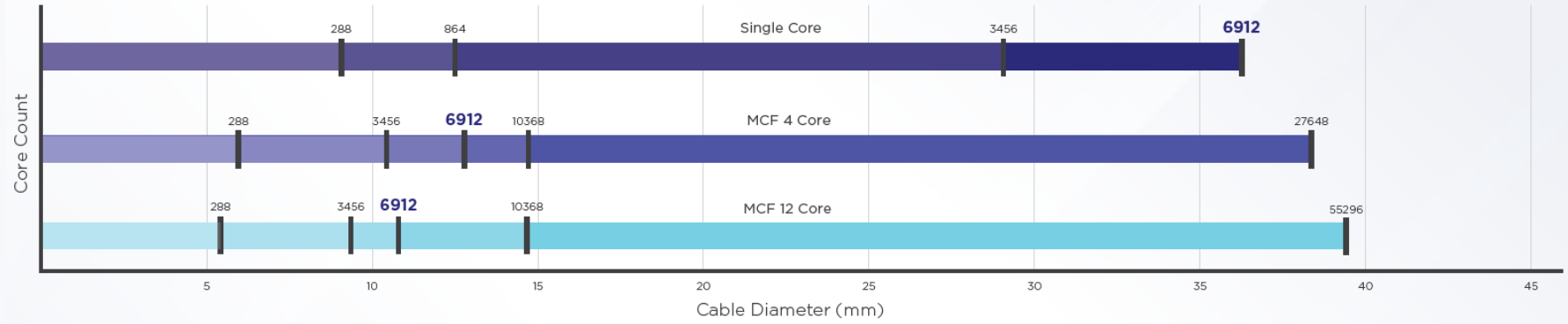
Digital transformations such as artificial intelligence (AI), machine learning, and the Internet of Things (IoT) are placing unprecedented demands on connected network infrastructure. Multi-core cables are the next step in responding to continuous and increasing demand for higher transmission capacity within limited duct, tray, and rack space.

6912 Cable Diameter Example:

Single-core fiber: OD ~37mm Area: 1075mm^2

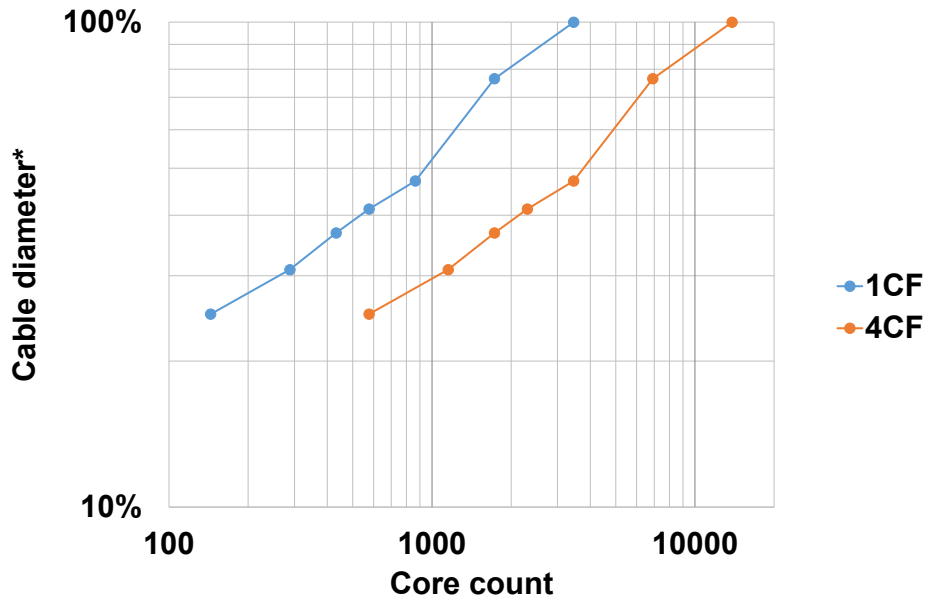
4-core multi-core fiber: OD ~13mm Area: 133mm^2

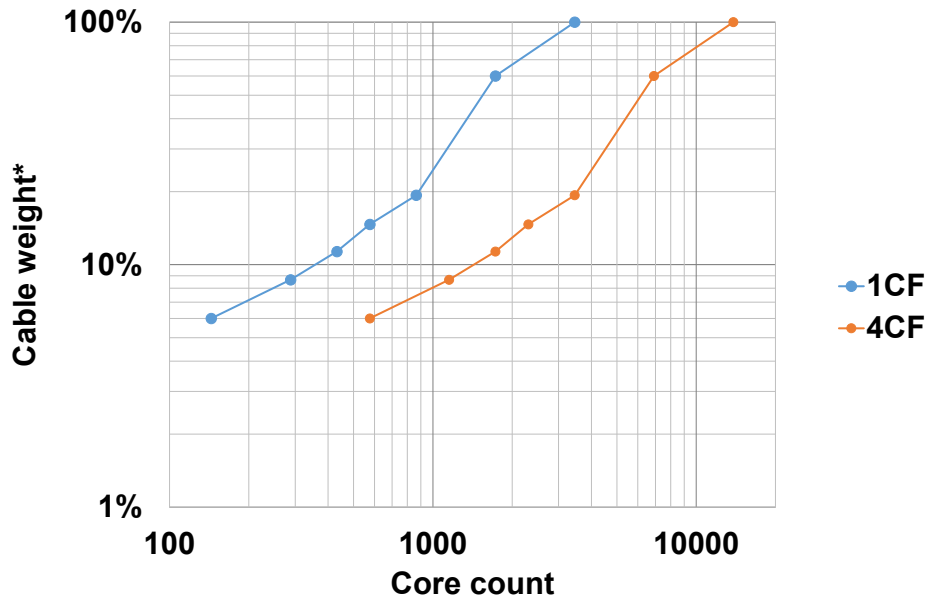
60-80% space savings in duct on a single-core to 4-core product comparison



A 6,912f cable is currently the industry's highest capacity commercially available optical fiber cable.

Multi-Core Fiber – Space & Weight Savings



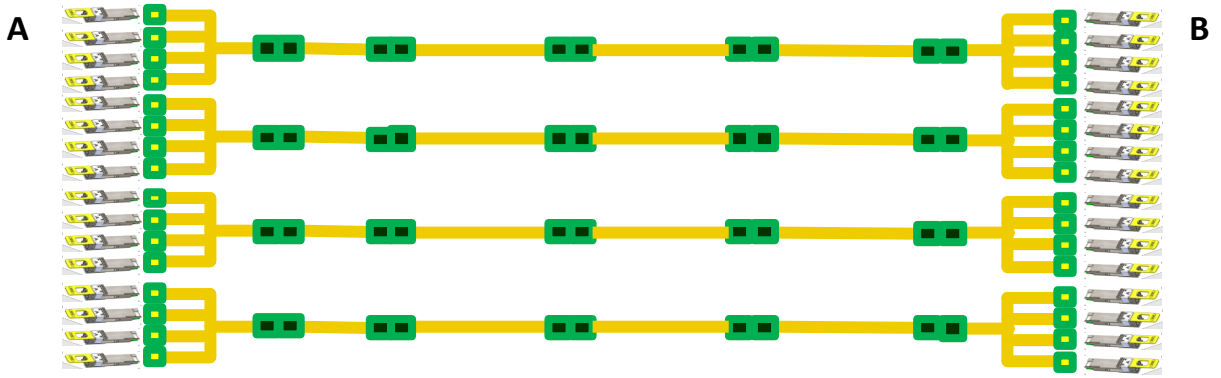


MCF significantly reduces cable size & weight per core

- Enables higher core count in a fixed-size duct
- Allows easier installation and reduces environmental load for the same core count

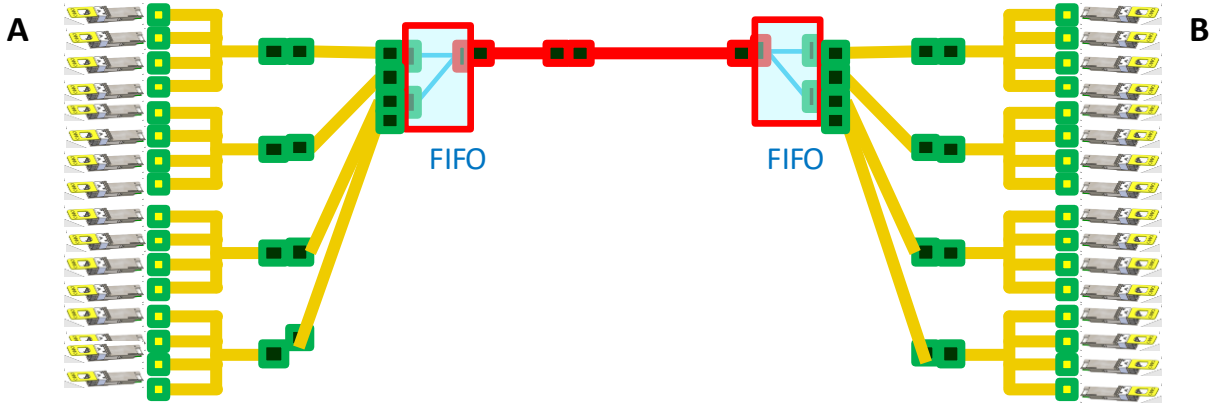
Single-Core Fiber

Example of a typical link with SCF in an AI DC: Transceiver links are collected into groups at the TOR and are transported via high-fiber-count trunks. Once they reach the end of the pathway, they branch out again to the transceiver at the other end.

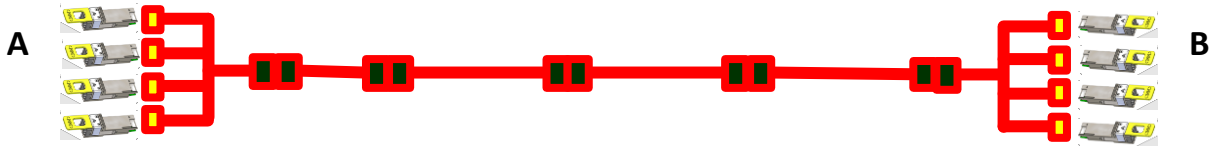


Multi-Core Fiber

As many fibers travel from point **A** to point **B** on the same path, MCF can collapse the trunks into a single cable. FIFO would be used close to the TOR to allow SCF to be used with traditional transceivers.



Ideally, MCF can be utilized over the entire pathway with MCF-native transceivers. This approach delivers the greatest value.

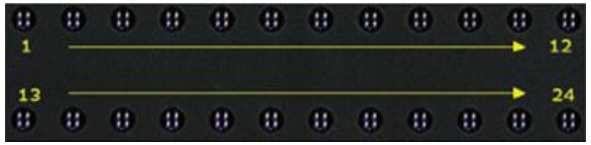


Multi-Core Connectors

EXAMPLE:

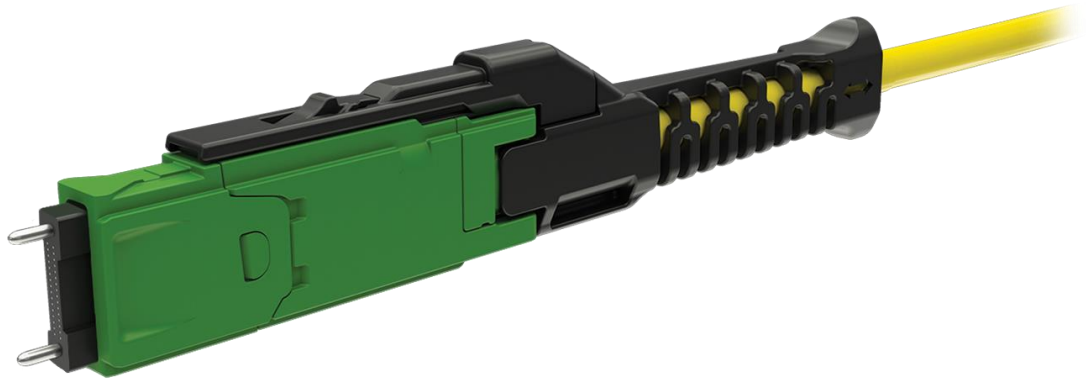
96-core MPO-APC connector using 4-core fiber





EXAMPLE:

64-core MMC-APC connector using 4-core fiber





Multi-Core Fiber – Challenges

Business Challenges

- More costly than single-core fiber per strand and connector; however, MCF's total cost of ownership is much better
- Damage during or after installation becomes more costly to repair
- Not standardized yet

Operational Challenges

Manufacturing

- Core alignment during drawing
- Limitation on fiber length compared to SCF
- Tolerance of core position
- Core alignment in connectors/splicing

Installation

- Field repair difficult
- Dust sensitivity – flat polish
- Not backward compatible with SCF

Network Design

- Requires FIFO unit to access individual cores
- Migration to native MCF transceivers and co-packaged optics integration (MCF to the ASIC)

Q&A

NEXT GENERATION THINKING™


SUMITOMO ELECTRIC
LIGHTWAVE